

How Does Chain-of-Thought Help Language Models Count? A Causal Test of Global Aggregation and Sequential Retrieval

Authors to be added

Working paper for an ICLR 2027 submission

Draft of July 29, 2026

Abstract

Counting sparse, semantically defined records in a long context requires every valid occurrence to affect the answer exactly once. In a realistic needle-in-a-haystack audit, native chain-of-thought (CoT), numbered enumeration, and bullet enumeration are substantially more accurate than direct non-thinking answers in the moderate-count regime. All interfaces degrade as count and context length grow, but current coefficients are not stable enough to support a universal empirical law.

We ask which computational change explains the CoT advantage. Rather than assuming that Direct maintains a noisy running counter, we test two Direct programs: record-wise running update and answer-time broad aggregation. We contrast both with a sequential CoT program consisting of targeted source retrieval, trace writing, semantic progress transition, next-source or stop control, and terminal readout. The experiment registers exact residual-stream token landmarks, operational attention-role scores, and length-preserving sibling stimuli that separate local updates from final totals. Representation claims require held-out decodability and the correct +1/0 dynamics; mechanism claims additionally require activation interventions to propagate through source identity, unique coverage, progress, terminal state, and the complete parsed answer. Qwen3-8B and Gemma4-E4B are co-primary mechanism models. PCA and attention maps remain descriptive until the corresponding causal links pass. Existing synthetic results establish measurement feasibility; realistic LLM mechanism results remain registered TODO experiments. **[TODO: Replace preliminary language with paired effect sizes and arrow-level recovery intervals after the locked causal study.]**

1 Introduction

Long-context retrieval benchmarks usually ask whether a model can recover one item from distractors. Counting is stricter: every valid record must contribute once, while invalid and repeated records must contribute zero. A correct answer therefore requires source discovery, duplicate-free coverage, aggregation, stopping, and numerical readout. Multi-needle tasks such as RULER expose failures that single-needle accuracy can conceal [4].

CoT often improves multi-step reasoning [12], but the emitted trace need not faithfully report the computation that produced the answer [9]. The question of this paper is therefore mechanistic:

How does CoT help a language model count, and which causal computation distinguishes it from a direct non-thinking answer?

We test a deliberately small contrast. Direct answering may update a predicate-conditioned state after each record, or it may construct the task-relevant total only near the answer through a broad multi-record aggregation route. CoT may instead issue a sequence of source-specific retrievals, write the retrieved items into an external trace, update semantic progress, and use that state to retrieve the next source or stop. The central candidate benefit is an algorithmic path change: a difficult high-fan-in retrieval-and-aggregation operation is replaced by repeated low-fan-in retrievals whose intermediate results remain available in the generated prefix. We do not assume in advance that either interface contains a scalar neural counter, that a visible index is faithful, or that more generated tokens are sufficient.

The paper follows one evidence chain:

$$\begin{aligned} \text{behavioral contrast} &\longrightarrow \text{registered computation} \longrightarrow \text{causal state} \\ &\longrightarrow \text{functional component} \longrightarrow \text{complete answer.} \end{aligned} \tag{1}$$

Each arrow has a separate falsification test. A probe or PCA curve can establish representation, not use. An attention heatmap can nominate a route, not establish its function. An ablation can show necessity, but the proposed explanation of CoT requires restoration of the intervening retrieval or count state and recovery of the complete parsed answer.

Our contributions are:

1. We consolidate the behavioral phenomenon over needle count, context length, response mode, and a broad model panel, while treating any parametric “law” as a held-out prediction problem rather than a fitted slogan.
2. We construct equal-length sibling families and prompt-policy controls that distinguish a Direct running update from terminal broad aggregation and remove visible-index leakage from CoT progress.
3. We register exact record, retrieval, progress, and terminal token landmarks, and define broad-aggregation, targeted-retrieval, and progress-transition candidates whose functional names require held-out interventions.
4. We register the decisive time-indexed chain: perturb routing, measure source identity and unique coverage, measure progress and terminal state, and restore a clean mediator while the upstream component remains perturbed.

The realistic mechanism study is co-primary in Qwen3-8B and Gemma4-E4B [2, 13]. Model identities, semantic sites, endpoints, and intervention logic are shared, but heads are aligned by function and relative depth rather than numerical index. Synthetic transformers serve as a controlled bridge for measurement design and training dynamics; they do not substitute for causal results in pretrained LLMs.

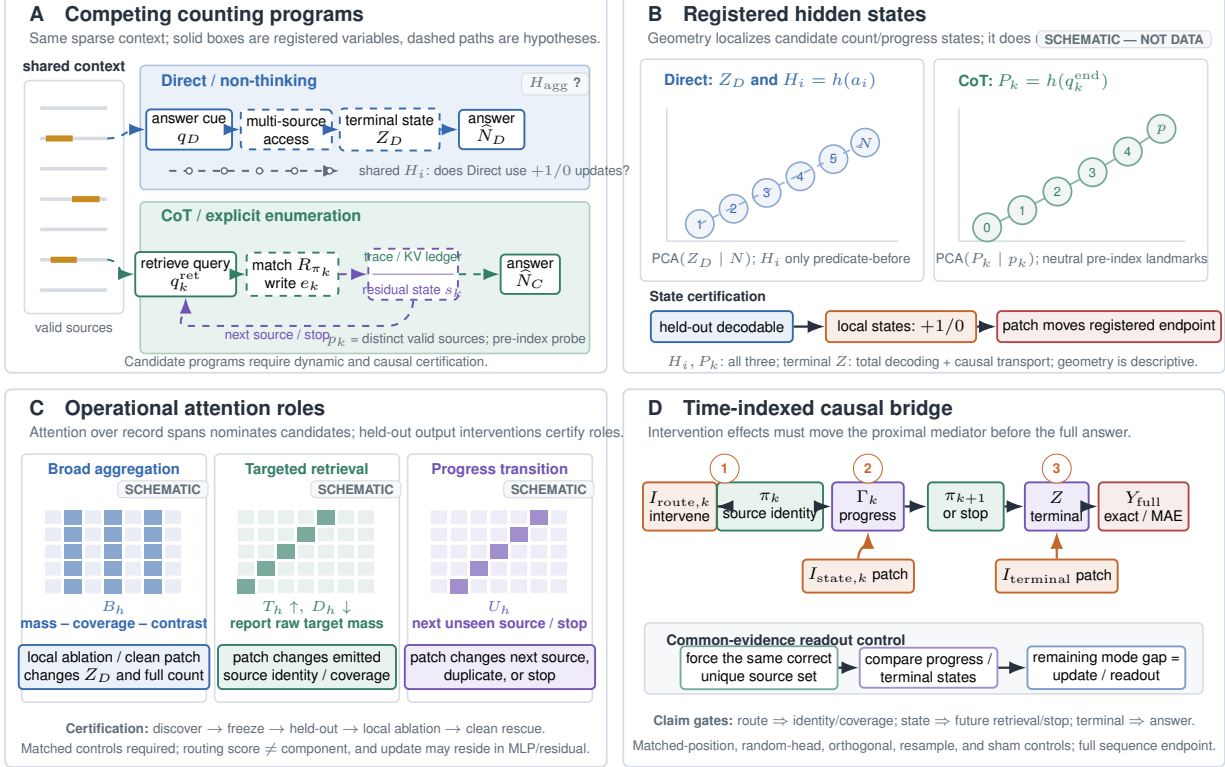


Figure 1: **Mechanism map and causal test plan.** **A:** Direct is tested for terminal multi-source aggregation and, only when the counting predicate precedes the records, for causal use of the shared record-end states H_i . CoT is tested for recurrent targeted-source retrieval, trace writing, semantic progress, and next-source or stop control. **B:** Registered hidden states separate Direct terminal Z_D , predicate-conditioned record-end H_i , CoT retrieval Q_k , and CoT progress P_k . Local counter claims require held-out decoding, correct $+1/0$ dynamics, and downstream transport; terminal states require total decoding and causal readout. **C:** Attention nominates broad-aggregation, targeted-retrieval, and progress-transition candidates; query-local ablation and clean patching assign function. **D:** A mechanism claim requires time-indexed effects on source identity/coverage, progress or stopping, terminal state, and the complete answer. The common-evidence control supplies both modes with the same correct source set to isolate state construction and readout. Miniature trajectories and heatmaps are analysis templates, not empirical results.

2 Experimental Design

2.1 Task and frozen naturalistic stimuli

A passage contains source occurrences $R_1, \dots, R_M$. Under a model-specific tokenizer, occurrence i occupies the half-open span $I_i = [s_i, e_i)$. A registered semantic predicate q gives

$$y_i = \psi_q(R_i) \in \{0, 1\}, \quad G_q = \{i : y_i = 1\}, \quad N = |G_q|. \quad (2)$$

The source identity is the *occurrence index* i , not merely a city or score string. Retrieving one occurrence twice is a duplicate; two distinct valid occurrences both contribute to the count.

The behavioral suite is the frozen Realistic NIAH V2 grid. Filler text is built from a content-deduplicated 218-source Paul Graham corpus. For each seed, the source documents are shuffled once and the five length conditions use nested prefixes; no essay is repeated to fill a context. We cross

$$L \in \{2000, 3000, 5000, 10000, 20000\}, \quad N \in \{1, 2, 3, 4, 5, 6, 8, 10, 20, 30\},$$

with seeds 1234, ..., 1243, yielding 500 frozen base passages. L is the exact post-insertion length under the frozen Qwen3-8B tokenizer; realized length under every evaluated tokenizer is also stored. Every valid record has the literal form

In the 2024 city score audit, <city> received a score of <score>.

Cities and scores are sampled without replacement within a passage. Candidate insertion sites are conservative sentence endings that exclude numeric strings, URLs, file paths, and common abbreviation periods. Sites are sampled without replacement when possible; any replacement or word-boundary fallback is logged. The generator changes only neutral filler and regenerates until the *post-insertion* canonical length is exact. A stimulus is rejected unless every target occurs exactly once, no target-like sentence occurs in the filler, every character span maps to a verified token span, and no insertion is truncated. Passage bytes, occurrence identities, insertion depths, and a SHA256 digest are frozen before expanding modes or models. Thus all modes and models for one behavioral stimulus receive the identical passage text.

2.2 Controlled sibling stimuli for mechanism identification

The naturalistic grid measures behavior but cannot by itself distinguish a record-wise counter from late aggregation: increasing N also inserts more record-like sentences. We therefore construct a disjoint, tokenizer-audited mechanism corpus. A stimulus family

$$\omega = (H, q, \{u_j, I_j, y_j\}_{j=1}^K) \tag{3}$$

contains the same filler H , $K = 10$ fixed record-like slots, opaque occurrence IDs u_j , and positive/negative variants that differ in exactly one predicate-critical attribute. Hard negatives use a wrong audit type, such as **city safety audit**, or a wrong document type, such as **city score report**. Year alone is never used as a negative because the frozen V2 predicate does not restrict year.

For each mechanism model, rejection sampling makes every positive/negative pair equal in token length. Slots are drawn once from ten normalized-depth strata at legal sentence boundaries, with a target separation of at least 128 model tokens. Slot text, order, and token positions remain fixed across siblings at one length. A balanced permutation chooses which N slots are positive, so validity is not predicted by depth, negative type, city, or score. Qwen and Gemma corpora are rendered separately to make post-insertion length exact under each tokenizer. They share the semantic family manifest, slot order, labels, and normalized depths, but are not assumed to share token IDs or absolute token indices.

We register two paired manipulations:

- A *prefix-shift* pair exchanges an early positive slot r and a late matched negative slot $s > r$, preserving final N , L , slot identities, and all downstream token indices while changing the cumulative count only for $r \leq j < s$.

- A *total-shift* pair flips one equal-length slot from negative to positive, producing $N \leftrightarrow N + 1$ without changing its location or the length of any suffix.

Every clean/corrupt pair stores complete prompt and continuation token IDs, modified character/token spans, and length-alignment checks.

TODO. D1—freeze the controlled mechanism corpus. Use $N \in \{0, \dots, 6, 8, 10\}$. At the primary length $L = 5k$, assign at least 120 independent families before activation capture: 60 discovery, 20 validation, and 40 locked test families. A family-cluster pilot power analysis may increase, but may not silently shrink, the locked set. After layers, heads, thresholds, and intervention doses are frozen, evaluate at least 40 new locked families at each of $L = 2k$ and $L = 10k$. Prefix shift is eligible only for $1 \leq N \leq K - 1$; at $N = 0$ or $N = K$, use the available one-direction total shift. Use data seeds disjoint from the behavioral grid and publish the semantic manifest, tokenized realizations, contamination audit, and all rejected generations.

2.3 Prompt serialization, reasoning policy, and decoding

The frozen behavioral layout is *predicate-before*, *suffix-after*. A common cue and the definition of a city-score audit record precede `<passage>`; the response request and mode-specific policy follow `</passage>`. The passage and final question are reused byte for byte across modes. The response policy and, where available, the model’s reasoning switch define the observable interface:

Table 1: Frozen response interfaces. Always-on reasoning checkpoints are not interpreted as non-thinking in a nominal Direct cell.

Mode	Response-policy suffix	Decoding	Primary role
Direct	One line, Total: <code><integer></code> ; no explanation or list	thinking off; greedy; 64 tokens	non-thinking endpoint
Bullet	Passage-order list of actual city: score pairs, then total	thinking off; greedy; 1536 tokens	aligned, index-free process control
Index	The same list with ordinary numeric indices, then total	thinking off; greedy; 1536 tokens	visible-index control
Native CoT	Concise native reasoning, followed by the same total line	thinking on; frozen model-specific policy; 4096 tokens	ecological reasoning condition

Because the mode-specific suffix follows the passage, a causal transformer cannot make passage-token states mode specific when the rendered prefix through the passage is identical. We verify this condition after applying each checkpoint’s exact chat template. Passing passage states are treated as a *shared scanning substrate*, not as evidence that CoT constructed a cleaner representation while reading. Mode-specific passage encoding is studied only in a separate *policy-before* control that moves the Direct/enumeration instruction before the passage; those states are never pooled with the frozen behavioral grid.

The mechanism study additionally crosses visible instruction (Direct versus unindexed enumeration) with the Qwen/Gemma reasoning switch (off versus on), forming a 2×2 instruction-by-policy factorial. Predicate placement has three levels: *predicate-before*, *predicate-after*, and *predicate-repeat*. Predicate-after record states cannot encode the task-specific cumulative count and provide

a causal-mask negative control; predicate-repeat distinguishes availability during encoding from instruction recency.

Data-generation and decoding seeds are separate fields. The ten V2 seeds are stimulus replicates, not repeated samples from one decoding distribution. For stochastic endpoints, clean, corrupt, and patched rollouts use common random numbers and at least three paired decoding seeds. Parse failure, format failure, and truncation remain in intent-to-treat summaries. We report teacher-forced likelihood of the *complete registered continuation* and free-generation exact count and MAE; a first-digit logit is never the sole endpoint. Equal-length null scratchpads control for extra tokens and recurrent computation.

TODO. D2—prompt and policy controls. Run the 2×2 instruction-by-policy factorial, the three predicate placements, and policy-before versus suffix-after on locked Qwen3-8B and Gemma4-E4B mechanism families. Freeze checkpoint and tokenizer revisions, chat templates, stop rules, generation budgets, and parsers. Hash every rendered prefix through the final passage token. Use the same sampler, maximum budget, and paired generation seeds within a reasoning-toggle contrast. Any claimed algorithmic change must survive controls for instruction recency, output length, null computation, and checkpoints that reason in every nominal mode.

2.4 Registered token landmarks and activation capture

For block ℓ , h_t^ℓ is the `resid_post` stream returned after that block’s residual additions and before block $\ell + 1$ consumes the state. Embedding states are diagnostic only. Every token index is obtained from character-to-token offsets on the *final model-specific chat-rendered sequence*; no index is inferred by adding the length of a standalone needle.

Controlled records end with the invariant delimiter

$$D_{\text{eor}} = [\text{END-RECORD}].$$

Writing a_i for the token overlapping its closing bracket, the registered Direct record state is $H_i^\ell = h_{a_i}^\ell$. The Direct terminal state $Z_D^\ell = h_{q_D}^\ell$ uses the token overlapping the colon in a teacher-forced **Total:** prefix, immediately before the first count token. In controlled CoT, the unindexed grammar is

$$\text{Item: } \langle \text{city} \rangle: \langle \text{score} \rangle [\text{END-RECORD}].$$

The retrieval state $Q_k^\ell = h_{q_k^{\text{ret}}}^\ell$ is extracted at the colon in **Item:**, before source-bearing content. The progress state $P_k^\ell = h_{q_k^{\text{end}}}^\ell$ is extracted at the closing bracket after the k th source mention and before any next index or label. The controlled terminal $Z_C^{\text{ctrl}, \ell}$ is the **Total:** colon after the canonical teacher-forced trace. A uniquely aligned **Total:** colon in a free-running native trace defines $Z_C^{\text{native}, \ell}$; native and controlled terminals are never pooled.

The chat prompt is tokenized with the exact generation-time template. A teacher-forced continuation is appended as continuation token IDs with no additional special tokens; we never retokenize the concatenated raw string in a way that can alter the final prompt token. A sample is rejected if the token overlapping **Item:** also overlaps the first source character or if the token overlapping **Total:** contains any count character. The manifest stores raw messages, rendered text, prompt and continuation token IDs, character spans, token spans, landmark indices, model revision, tokenizer revision, and hook name.

At a Direct record endpoint, the semantic label is

$$c_q(a_i) = \sum_{j \leq i} y_j. \quad (4)$$

For a CoT trace, align emitted step k to $\pi_k \in \{1, \dots, M\} \cup \{\perp\}$, where $\perp$ denotes an unaligned item. Define

$$\mathcal{V}_k = \text{uniq}\{\pi_u : 1 \leq u \leq k, \pi_u \neq \perp\}, \quad p_k = |\mathcal{V}_k \cap G_q|. \quad (5)$$

Thus CoT progress is the number of distinct valid occurrences actually retrieved, not the visible step number.

Table 2: Registered residual-stream landmarks. Primary state analyses use one token per landmark, not a record-span average.

State	Exact token	Registered label and use
Direct local H_i^ℓ	Closing bracket of D_{eor} after slot i	$c_q(a_i)$; +1/0 dynamics and downstream propagation, only when the predicate is already available
Direct terminal Z_D^ℓ	Colon in teacher-forced Total :	Total N ; terminal aggregation and complete-answer readout
CoT retrieval Q_k^ℓ	Colon in unindexed Item :	Gold next source identity; targeted retrieval
CoT progress P_k^ℓ	Closing bracket after the k th emitted source	Semantic progress p_k ; next-source and continue/stop control
Controlled CoT terminal $Z_C^{\text{ctrl}, \ell}$	Colon in Total : after the canonical forced trace	Terminal readout given registered evidence
Native CoT terminal $Z_C^{\text{native}, \ell}$	Uniquely parsed Total : colon in a natural trace	Ecological replication; never pooled with the controlled terminal

2.5 Analysis unit, splitting, and averaging

The atomic representation observation is

one family $\times$ one rendered variant $\times$ one registered landmark $\times$ one layer.

The stimulus family is the independent unit for splitting and uncertainty. All siblings, modes, predicate placements, (N, L) variants, and clean/corrupt counterfactuals remain in the same family split. Discovery fits preprocessing and selects layers/heads; validation freezes hyperparameters, thresholds, and intervention doses; locked test families are opened once. Additional transfer tests hold out record lexemes, templates, and complete (N, L) cells.

Let $\mathcal{B}$ be families, $\mathcal{V}_b$ the eligible rendered variants of family b , and J_{bv} the eligible landmarks in variant v . Representation objectives use

$$\frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} \frac{1}{|\mathcal{V}_b|} \sum_{v \in \mathcal{V}_b} \frac{1}{|J_{bv}|} \sum_{j \in J_{bv}} \mathcal{L}(g(h_{bvj}^\ell), y_{bvj}). \quad (6)$$

No hidden state is averaged across prompts before PCA, probing, or primary patching. PCA centering, nuisance regression, decoder regularization, and directions are fitted on discovery observations only. Figures may show individual locked-test points and family centroids, but centroids are summaries rather than substituted observations.

Primary natural-state patching uses one matched donor. A discovery-set mean state is only an ablation control. Geometric steering is the sole causal intervention whose direction is defined by a cross-prompt mean, estimated from paired discovery-family differences. Attention statistics are averaged first over eligible steps within a variant, then over variants within a family, and finally equally over families. Raw hidden vectors are never averaged across layers, checkpoints, token roles, or model families.

3 Behavioral Phenomenon and Empirical Response Surfaces

3.1 Outcomes and paired estimands

The task, occurrence identity, response interfaces, and passage construction are defined in [section 2](#). Direct versus Native CoT is the primary *interface contrast*: in the frozen V2 suite it changes both the visible response policy and the model’s reasoning policy, so it is not interpreted as a single-variable causal effect of CoT. Bullet and Index separately test an explicit evidence ledger without unrestricted native reasoning.

The primary outcome is intent-to-treat numerical exactness of the parsed integer. *Exact-count accuracy* means that the final valid **Total: <integer>** equals N . *Registered success* additionally requires the mode-specific output format and no length truncation. We report both and never call registered success “accuracy” without qualification. Parse failure, strict format failure, truncation, protocol failure, trace precision/recall, duplicate rate, and final-versus-enumerated consistency are separate outcomes.

For model r , mode m , target count N , and length L , define

$$A_{rm}(N, L) = \Pr(\hat{C} = C \mid r, m, N, L), \quad e = \hat{C} - C, \quad D_{rm}(N, L) = \mathbb{E}[|e| \mid r, m, N, L]. \quad (7)$$

The paired interface contrast is

$$\Delta_r(N, L) = A_{r, \text{Native}}(N, L) - A_{r, \text{Direct}}(N, L). \quad (8)$$

All mode effects are paired by frozen stimulus ID. Numerical exactness is primary; signed bias and absolute error diagnose direction and magnitude.

For a trace, let $\hat{G}$ be the emitted multiset of aligned occurrence IDs and let $\hat{N}$ be the parsed total. When both are available,

$$\hat{N} - N = \underbrace{\text{FP} - \text{FN} + \text{Dup}}_{\text{retrieval and coverage}} + \underbrace{(\hat{N} - |\hat{G}|)}_{\text{aggregation or protocol}}, \quad (9)$$

where

$$\text{FP} = |\text{uniq}(\hat{G}) \setminus G_q|, \quad \text{FN} = |G_q \setminus \text{uniq}(\hat{G})|, \quad \text{Dup} = |\hat{G}| - |\text{uniq}(\hat{G})|.$$

This identity is applied only when the total and source identities are parseable. Every other request remains an intent-to-treat parse, alignment, truncation, or protocol failure rather than being dropped from a denominator.

3.2 The behavioral model panel

The empirical response-surface panel is intentionally broader than the mechanism panel. It contains a within-family Qwen size series and several cross-family reasoning models:

Table 3: Frozen behavioral panel and its role. Gemma4-E4B has approximately 4.5B effective/non-embedding parameters and 8B parameters when embeddings are included; we therefore report both conventions and do not treat it as a dense 4B point.

Role	Models	Use and restriction
Within-family size panel	Qwen3-4B, 8B, 14B, 32B	Develop candidate surfaces on 4B/8B/32B; use 14B as a fully held-out interpolation test. Do not refit the functional form after seeing 14B.
Cross-family behavior	Gemma4-E4B; GLM-4-9B and GLM-Z1-9B; Cogito-v1-Preview-8B; Nemotron-Nano-v2-9B; Nemotron-3-Nano-4B	Test direction and heterogeneity of the N, L effects. Do not pool their parameter counts into one scaling coefficient.
Co-primary mechanism	Qwen3-8B and Gemma4-E4B	Direct versus native CoT is primary; Index is a structured positive control. Roles transfer by semantic site and causal function, not head number.

The current focused analysis contains eight logical slots: Qwen3-4B/8B/32B, Gemma4-E4B, one GLM slot, Cogito-v1-Preview-8B, and the two Nemotron models. Qwen3-14B is absent and is a confirmatory TODO. The GLM slot is stitched: GLM-4-9B supplies Direct/Index/Bullet, whereas GLM-Z1-9B supplies CoT. It is useful descriptively but is not a same-weight thinking toggle. Models outside this panel are retained in the appendix with numerical accuracy and an explicit inclusion failure, rather than being silently dropped.

3.3 Preliminary core-regime result

The current realistic audit contains 53 model-mode cells and 500 requests per complete cell, for 26,500 request rows. The grid is

$$N \in \{1, 2, 3, 4, 5, 6, 8, 10, 20, 30\}, \quad L \in \{2, 3, 5, 10, 20\} \text{k tokens},$$

with ten context seeds per (N, L) cell. We define the *core regime* before analysis as $N \leq 10$ and $L \leq 10\text{k}$, giving 320 requests per complete model-mode cell.

This table is a preliminary behavioral summary, not yet a homogeneous frozen panel. The current report combines formal-V2 Direct/Native rows with v2.1 enumeration replacements and contains checkpoint-specific protocol failures. Every final row must therefore retain `source_version`, checkpoint and tokenizer revisions, rendered-template hash, decoding policy, and request-level failure flags. Aggregating across those fields before the audit in TODO B0 would turn provenance differences into apparent mode effects.

Table 4: Preliminary numerical exact accuracy in the core regime $N \leq 10, L \leq 10k$ (320 requests per available cell). The table reports behavior, not a same-checkpoint causal mode effect. Qwen3-14B has not yet been run.

Model / logical slot	Direct	Index	Bullet	CoT
Qwen3-4B	.519	.934	.919	.988
Qwen3-8B	.716	.991	.991	.997
Qwen3-14B	<i>TODO: all four modes</i>			
Qwen3-32B	.781	.994	.997	.994
Gemma4-E4B	.303	.966	.941	.922
GLM-4 / GLM-Z1 [†]	.381	.853	.844	.834
Cogito-v1-Preview-8B	.506	.625	.938	.897
Nemotron-Nano-v2-9B	.322	.825	.738	.844
Nemotron-3-Nano-4B	.478	.797	.619	.794

[†]Direct/Index/Bullet use GLM-4-9B; CoT uses GLM-Z1-9B. The row is excluded from same-weight mode-effect estimates.

Three empirical patterns motivate the mechanism study. First, all available trace-generating cells in [table 4](#) exceed their Direct counterpart. Second, CoT, Index, and Bullet occupy a similarly high band for the reliable Qwen, Gemma, and GLM cells; deviations such as Cogito Index and Nemotron Bullet show that trace format is not interchangeable for every model. Third, Direct is already accurate on some easy cells, which permits paired success/failure and non-ceiling mechanism analyses rather than comparing only a successful CoT state with a failed Direct state.

Across the full grid, accuracy generally decreases with larger N and L , while $|e|$ generally increases. The trend is not pointwise monotone for every model and mode. For example, parsing and truncation can distort Gemma4-E4B Direct length summaries, and the shared Direct response form performs poorly for Qwen3-8B. We therefore state the trend qualitatively until the held-out tests below are complete. The mechanism paper does not depend on a universal scaling law.

TODO. B0—configuration, provenance, and failure audit. For every requested checkpoint and mode, verify the frozen stimulus SHA256, checkpoint/tokenizer revision, chat-template revision, reasoning policy, generation budget, parser version, and request-level `source_version`. Report all attempted requests, including parse, format, truncation, answer-leakage, invalid-mode, and protocol failures. Keep formal V2 and v2.1 replacement rows distinct until the intended homogeneous panel is regenerated. Record a frozen exclusion code for every model/mode that cannot enter the main comparison.

TODO. B1—paired behavioral grid. Run Direct, Index, Bullet, and Native CoT on the same 500 frozen stimuli for the full requested model panel. For Qwen3-8B and Gemma4-E4B, additionally cross response mode with deterministic matched decoding, each model’s documented native policy, and a fixed generated-token budget. Report intent-to-treat exact count, registered success, conditional numerical accuracy, signed error, MAE, the decomposition in [equation \(9\)](#), parse/format/truncation failures, and family-cluster intervals. Predeclare overlapping-difficulty cells for Gemma because Direct is near floor. If the numerical interface gap disappears under policy and budget matching, do not advance a reasoning-policy explanation.

3.4 Response surfaces are descriptive unless they predict held-out models

We use one generic family of response surfaces for both accuracy and error:

$$\text{logit } A_{rm}(N, L) = \alpha_{rm}^{(A)} + \beta_{N,rm}^{(A)} f(N) + \beta_{L,rm}^{(A)} g(L) + \beta_{NL,rm}^{(A)} f(N)g(L), \quad (10)$$

$$\log(1 + D_{rm}(N, L)) = \alpha_{rm}^{(D)} + \beta_{N,rm}^{(D)} f(N) + \beta_{L,rm}^{(D)} g(L) + \beta_{NL,rm}^{(D)} f(N)g(L). \quad (11)$$

Candidate bases $f(N) \in \{N, \log N\}$ and $g(L) \in \{L/1000, \log(L/1000)\}$, interactions, regularization, and held-condition splits are frozen before Qwen3-14B is generated. Coefficients are partially pooled within mode but remain model specific. A parametric form is promoted to an empirical law only if it predicts the held-out 14B results better than a no-size hierarchical baseline and remains stable under leave-one-size-out and family-held-out checks.

Current evidence does not meet that standard. The Native $N \times L$ interaction is the most stable descriptive term across the eight development slots. Direct has useful average predictive structure but unstable coefficients and a negative cross-validated R^2 for Qwen3-8B under the shared form. Index and Bullet have weak or unreliable shared surfaces. Accordingly, the main text will emphasize nonparametric (N, L) heatmaps and partial contrasts; full coefficients and failed fits belong in the appendix.

TODO. B2—held-out response-surface test and presentation. Using only Qwen3-4B/8B/32B, freeze the candidate basis, partial-pooling model, and all splits. Generate Qwen3-14B on the identical 50 (N, L) cells and ten seeds in all four modes (2,000 requests). Report held-out log score, calibration, accuracy error, and absolute-error prediction against a no-size baseline; do not reselect after observing 14B. The main behavior figure should combine (a) Qwen size-panel predictions with the untouched 14B points and (b) Direct-versus-CoT cross-model partial-effect intervals. If validation succeeds, preregister a second extrapolation test with additional small Qwen checkpoints; if it fails, do not expand the size panel, replace “law” with qualitative response surfaces, and retain coefficients only as descriptive summaries.

3.5 Behavior and mechanism panels remain separate

The behavior panel is deliberately broad, whereas causal mechanism analysis uses only Qwen3-8B and Gemma4-E4B on the controlled sibling corpus in [section 2](#). Native CoT is the ecological interface; unindexed enumeration supplies the primary aligned trace for landmark discovery and causal intervention; Index is the visible-label control. Canonical teacher-forced traces measure local competence, while free-running traces measure execution. Every emitted item is aligned to an occurrence or labeled duplicate, invalid, hallucinated, or unresolved. Primary analyses include all paired locked families; both-correct and error strata are diagnostics rather than selection criteria.

4 Competing Computations and Registered Count States

4.1 Two Direct hypotheses and one sequential CoT program

We use minimal equations to specify what each experiment must distinguish, without assuming how a transformer implements the operation. For Direct,

$$w_i^D = \mathcal{R}_D(q, R_i), \quad Z_D = \mathcal{G}_D(\{w_i^D\}_{i=1}^M), \quad \hat{N}_D = \mathcal{Q}_D(Z_D). \quad (12)$$

$\mathcal{R}_D$ routes record information, $\mathcal{G}_D$ constructs a terminal task state, and $\mathcal{Q}_D$ reads out the count. This notation supports two competing hypotheses:

H_{run} : record-wise update. A predicate-conditioned record state is updated by approximately +1 after a valid occurrence and by 0 after a matched non-target, and the updated state affects later computation.

H_{agg} : terminal aggregation. Local states may contain content or ordinal information, but the task-relevant total is constructed mainly near **Total**: through broad multi-record access and terminal readout. The alternatives are not defined by whether a probe decodes N . They are distinguished by when a controlled +1 becomes available and where a natural activation patch transports it.

For CoT or explicit enumeration,

$$\begin{aligned}\pi_k &= \Pi_C(q, \Gamma_{k-1}), & e_k &= \mathcal{W}_C(R_{\pi_k}), \\ \Gamma_k &= \mathcal{F}_C(\Gamma_{k-1}, e_k), & \hat{N}_C &= \mathcal{Q}_C(\Gamma_K).\end{aligned}\tag{13}$$

π_k is the source occurrence actually selected at step k , e_k is the written evidence, and Γ_k is the progress carrier used to choose another source or stop. Γ_k may combine residual state P_k , the visible trace, and its KV cache; it is not assumed to be a scalar neural counter. The registered semantic label is p_k from [equation \(5\)](#), not the visible index k .

4.2 When we use the word counter

A registered state is called a *counter* only after three nested tests:

1. **Decodable.** A decoder trained on disjoint families predicts the registered cumulative count, semantic progress, or final total beyond nuisance-only and token/position controls.
2. **Dynamically consistent.** Direct record states exhibit the sibling-controlled +1/0 profile; CoT progress increases after a newly retrieved valid occurrence and not after a matched non-target or repeated occurrence.
3. **Causally used.** A single-donor natural patch or a validation-selected steering intervention moves a registered downstream endpoint in the predicted direction.

A state that passes only the first test is a count-decodable representation. PCA is a visualization and satisfies none of these criteria by itself. Terminal states Z_D and Z_C are not called running counters; they require total decodability and causal transport to the complete answer.

4.3 Held-out decoding and geometry

The token landmarks and family-equal analysis unit are fixed in [table 2](#) and [equation \(6\)](#). At each layer and semantic role, a nuisance-only model $g_0(u)$ is compared with a state-augmented model $g_1(h^\ell, u)$. For Direct local states,

$$u_D = (i, K, N, L, a_i/L, N/L, \text{depth bin, template ID});$$

for CoT progress,

$$u_C = (k, N, L, \text{trace-token length, visible-label identity}).$$

Remaining count $N - c_q(a_i)$ is a separate outcome, not placed in the same nuisance vector as both N and $c_q(a_i)$. Evidence for a state is the held-out improvement of g_1 over g_0 , not raw predictability of a count correlated with position or density.

For visualization, one shared PCA basis per model and layer is fitted on a discovery sample balanced over registered role, interface, semantic label, and (N, L) cell; the downsampling seed is stored. The frozen basis projects individual locked-test states. Mode-specific PCA is a sensitivity analysis. Correct and incorrect outputs are both included in the primary fit and may be stratified only for a registered diagnostic. No trajectory is created by averaging states within a prompt or by selecting successful traces.

TODO Figure: Registered states and causal transport		
<i>Held-out geometry</i>	<i>Dynamic tests</i>	<i>Causal transport</i>
individual H_i, P_k, Z_D, Z_C	Direct prefix-shift 0/1/0	single-donor patching
shared PCA basis	CoT new/invalid/repeated	paired geometric steering
family centroids as summaries	predicate-after control	complete sequence endpoint
Qwen3-8B and Gemma4-E4B use the same registered variables but independently selected layers.		
Cross-role dispersion is descriptive only.		

Figure 2: Planned representation figure. Readability, correct dynamics, and causal transport are shown separately; only their conjunction licenses a counter claim.

TODO. R1—registered-state decoding. For every state in [table 2](#), fit family-weighted nuisance-only and state-augmented decoders for Qwen3-8B and Gemma4-E4B. Freeze pre-processing, ridge penalties, PCA bases, and layer selection before the locked test. Report incremental held-out R^2 /MAE, unseen identity and template generalization, (N, L) transfer, conditional dispersion, and correct/error strata. Plot individual locked-test points plus family centroids; do not fit a projection to the presentation prompts.

4.4 Direct: running update or terminal aggregation?

For predicate-before and predicate-repeat prompts, extract H_i^ℓ at all K controlled slots, not only positives. The primary dynamic test uses prefix-shift siblings A and B that exchange a positive at r and a matched negative at $s > r$. Let $\tilde{H}_j^{A,\ell}$ and $\tilde{H}_j^{B,\ell}$ be nuisance-residualized states at the same unchanged landmark, and let g_h^ℓ be the hidden-only decoder frozen on discovery families. Define

$$\delta_j^\ell = g_h^\ell(\tilde{H}_j^{A,\ell}) - g_h^\ell(\tilde{H}_j^{B,\ell}). \quad (14)$$

H_{run} predicts

$$\delta_j^\ell \simeq \begin{cases} 0, & j < r, \\ 1, & r \leq j < s, \\ 0, & j \geq s. \end{cases}$$

This same-position contrast fixes final N , record identity, local token, length, and location. Adjacent-position increments inside one prompt are secondary content/update diagnostics. Predicate-after states provide the causal-mask negative control.

For causal transport, a total-shift pair differs at one equal-length slot r , but the primary patch is applied at the first *unchanged* downstream delimiter $j > r$. Its local token and suffix are identical across siblings, reducing record-content transport. Patching the modified record’s own endpoint is retained only as a positive evidence-transport control. Endpoints are decoded cumulative count at later landmarks, Z_D , complete continuation likelihood after **Total:**, and free-generated exact count/MAE. Identity-preserving positive-to-positive patches, matched negative-to-negative patches, unrelated positions, random states, norm-matched states, and sham patches are controls.

H_{agg} also requires positive evidence; it is not accepted merely because the local test fails. We test whether query-local multi-record routes carry N into Z_D , whether total-shift terminal patches move the complete answer, and whether those effects survive controls for local ordinal information. Evidence for H_{run} requires decodability, the 0/1/0 dynamic profile, and propagated downstream patches. Evidence for H_{agg} requires a selectively causal terminal route/state intervention.

TODO. R2—Direct dynamics and patches. At every frozen candidate layer, run prefix-shift and total-shift experiments on locked families. Use the same-position hidden-only contrast at unchanged landmarks for the primary +1/0 test; do not treat a second independent valid occurrence as a no-op. Patch single natural donors at unchanged downstream H_j and at Z_D separately. Measure later cumulative-state decoding, terminal state, complete-answer likelihood, and free-generation exact count/MAE. Include predicate-after and predicate-repeat, identity-preserving, position, random, norm, and sham controls. Report separate evidence for H_{run} and H_{agg} ; a failed local counter test is not by itself evidence for aggregation.

4.5 CoT progress without label leakage

At step k , let the registered next source in passage order be

$$g_k = \arg \min_{i \in G_q \setminus \mathcal{V}_{k-1}} s_i, \quad (15)$$

where s_i is source position. If the set is empty, the target is STOP. Gold-target alignment is the primary estimand; attention to the source the model actually emits, π_k , is reported separately as execution alignment so a consistently wrong retrieval cannot receive a high mechanism score.

For each visible step k , we teacher-force three equal-length prefixes whose newest item is a new gold source ($\Delta p = 1$), a matched non-target ($\Delta p = 0$), or a repeat of an already visited occurrence ($\Delta p = 0$). Complementary prefixes hold p_k fixed while changing trace extent. Numeric, random nonnumeric, permuted, and absent labels appear outside the invariant **Item:** and D_{eor} tokens. A state that tracks p_k under these transformations is evidence for semantic progress; a state that tracks only k or the displayed label is scaffolding.

A clean P_k^ℓ is patched into a matched corrupt prefix before the model chooses its next source. Proximal endpoints are the probability and generated identity of g_{k+1} , duplicate probability, and continue/stop probability; later endpoints are unique coverage, P_{k+1} , terminal state, and the complete count. Residual-only transport patches P_k^ℓ while preserving the receiver trace/KV cache. Trace/KV-only path patching transports donor trace keys and values while preserving the receiver residual. These carriers are complementary and are not conflated into one scalar counter.

For geometric steering, the discovery direction is the normalized paired difference

$$v_\ell = \frac{\mathbb{E}[P_k^\ell \mid \Delta p = 1] - \mathbb{E}[P_k^\ell \mid \Delta p = 0]}{\|\mathbb{E}[P_k^\ell \mid \Delta p = 1] - \mathbb{E}[P_k^\ell \mid \Delta p = 0]\|_2}. \quad (16)$$

The layer and dose are selected on validation families. Norm-matched random and orthogonal directions are controls. Steering defines one direction from a cross-prompt mean; primary patching still uses a single matched natural donor.

TODO. R3—CoT anti-leakage and state control. Run the equal-length new-gold/non-target/repeated-source prefix experiment with unindexed, random-label, and permuted-index interfaces. Test same- k different- p_k and same- p_k different-extent pairs. Separately transport residual state, trace/KV state, and both; run single-donor natural patches and a validation-selected steering sweep. Report gold next-source likelihood and identity, emitted-source alignment, duplicate and stop behavior, later progress, unique coverage, and the complete parsed count. Require effects beyond norm-matched, random, orthogonal, position, and sham controls before calling P_k a semantic progress carrier.

4.6 Conditional state dispersion is not probe error

Let $\tilde{h}^\ell$ be the residual after a discovery-set nuisance model removes registered position, length, density, template, and label covariates. For a semantic count or progress value c , define family-equally weighted moments

$$\mu_c^\ell = \mathbb{E}[\tilde{h}^\ell \mid c], \quad \Sigma_c^\ell = \text{Cov}(\tilde{h}^\ell \mid c).$$

For adjacent values, let

$$d_c = \mu_{c+1}^\ell - \mu_c^\ell, \quad u_c = \frac{d_c}{\|d_c\|_2 + \epsilon}, \quad \bar{\Sigma}_c = \frac{\Sigma_c^\ell + \Sigma_{c+1}^\ell}{2}.$$

The registered local confusion statistic is

$$\nu_{\text{state}}^\ell(c) = \frac{u_c^\top \bar{\Sigma}_c u_c}{\|d_c\|_2^2 + \epsilon}. \quad (17)$$

It is reported only when adjacent separation exceeds a validation-frozen threshold. Held-out decoder error measures accessibility to a chosen decoder and is reported separately. We test whether Direct terminal dispersion or error grows with N, L , and whether controlled CoT terminal states are less confusable. Because H_i, P_k , and Z occupy different token roles, cross-role dispersion comparisons are exploratory and cannot establish an algorithmic path change.

5 Routing and Update Components

5.1 Three operational attention scores

The state analysis asks what is represented. Attention is used only to nominate components that may route information into those states. Let $A_{q,t}^{\ell h}$ be attention from registered query token q to token t in layer ℓ , head h . For the complete preregistered span I_i of source occurrence i , define

$$a_i^{\ell h}(q) = \sum_{t \in I_i} A_{q,t}^{\ell h}. \quad (18)$$

We sum over the complete span rather than selecting a favorable token after inspection. Every report includes raw span mass, span-density $a_i^{\ell h}(q)/|I_i|$, and enrichment over a uniform-attention baseline, so long source strings do not receive an unqualified advantage.

Broad aggregation candidate. At the Direct answer cue q_D , let

$$M_G^{\ell h} = \sum_{i \in G_q} a_i^{\ell h}(q_D), \quad r_i^{\ell h} = \frac{a_i^{\ell h}(q_D)}{M_G^{\ell h} + \epsilon}.$$

For $N \geq 2$, define

$$B_{\ell h} = \mathbb{E}_\omega \left[M_G^{\ell h} \frac{\exp\left(-\sum_{i \in G_q} r_i^{\ell h} \log(r_i^{\ell h} + \epsilon)\right)}{N} \right]. \quad (19)$$

The second factor is effective source coverage divided by N : it is one for uniform mass across all gold occurrences and $1/N$ when mass is concentrated on one. We also register the gold-versus-matched-negative contrast

$$C_{\ell h}^B = \mathbb{E}_\omega \left[\frac{1}{N} \sum_{i \in G_q} a_i^{\ell h}(q_D) - \frac{1}{|G_q^-|} \sum_{i \in G_q^-} a_i^{\ell h}(q_D) \right]. \quad (20)$$

A large $B_{\ell h}$ without positive $C_{\ell h}^B$ is nonspecific broad access. A head or bundle earns the *broad aggregation component* name only if its query-local output intervention changes Z_D or the complete count and a clean patch restores that endpoint.

Targeted retrieval candidate. At CoT retrieval step k , the registered gold source g_k is defined by [equation \(15\)](#); $\mathcal{V}_{k-1}$ is the set already visited. Define

$$T_{\ell h} = \mathbb{E}_\omega \mathbb{E}_{k|\omega} \left[\frac{a_{g_k}^{\ell h}(q_k^{\text{ret}})}{\sum_{i=1}^K a_i^{\ell h}(q_k^{\text{ret}}) + \epsilon} \right], \quad (21)$$

$$D_{\ell h} = \mathbb{E}_\omega \mathbb{E}_{k|\omega} \left[\frac{\sum_{i \in \mathcal{V}_{k-1}} a_i^{\ell h}(q_k^{\text{ret}})}{\sum_{i=1}^K a_i^{\ell h}(q_k^{\text{ret}}) + \epsilon} \right]. \quad (22)$$

We report raw a_{g_k} , $T_{\ell h}$, $D_{\ell h}$, source-span top-1, and the gold-versus-best-wrong QK margin. Attention to the source the model actually emits, π_k , is an execution-alignment statistic and never replaces g_k in the candidate score. A targeted retrieval component must have high gold-target mass, low visited-source mass, and a causal effect on the emitted source identity or source-identity margin.

Progress-transition candidate. For a nonterminal controlled prefix, the next registered source is g_{k+1} . At the progress landmark q_k^{end} , define

$$U_{\ell h} = \mathbb{E}_\omega \mathbb{E}_{k|\omega} \left[\frac{a_{g_{k+1}}^{\ell h}(q_k^{\text{end}})}{\sum_{i=1}^K a_i^{\ell h}(q_k^{\text{end}}) + \epsilon} \right]. \quad (23)$$

Terminal prefixes are scored by the probability of **Total**:/stop and by free-running stopping behavior rather than by forcing a nonexistent source target. We use *progress-transition component* or *next-source control*; “successor head” is reserved for a component whose token-independent OV map survives relabeling and implements an abstract $k \mapsto k + 1$ operation [3].

Table 5: Operational component roles. Scores nominate candidates; the final column determines whether a functional name is licensed.

Role	Registered query	Discovery evidence	Proximal causal endpoint
Broad aggregation	Direct q_D	$B_{\ell h}$, $C_{\ell h}^B$, span-density sensitivity	Z_D , complete numeric-string likelihood, exact count
Targeted retrieval	CoT q_k^{ret}	$T_{\ell h}$, low $D_{\ell h}$, gold/wrong QK margin	gold/emitted source identity and unique coverage
Progress transition	CoT q_k^{end}	$U_{\ell h}$, terminal stop diagnostics	next source, repeat, continue, or stop

5.2 Discovery, freezing, and family-equal aggregation

Scores are computed at each eligible query, averaged over steps within a rendered variant, then over variants within a family, and only then equally over families. Discovery ranks candidates; validation freezes layer, bundle size, score thresholds, and all causal endpoints. A family-wise max-statistic permutation screen limits false discovery. We preregister at most five candidates per role and model for the locked causal test and apply Holm correction across the frozen role-by-endpoint tests, not across heatmaps selected after inspection.

Attention maps are retained as interpretable summaries, but they do not establish information flow. Ablations are applied to a head’s contribution after value aggregation and before/through output projection at the registered query, not globally when a local intervention is available. Controls include same-layer random heads, attention-mass-matched heads, wrong query rows, lexically adjacent tokens, unrelated source spans, resample and mean ablation, and clean-to-clean and corrupt-to-corrupt sham patches.

TODO Figure: Functional routing components		
<i>Broad aggregation</i>	<i>Targeted retrieval</i>	<i>Progress transition</i>
Direct answer-to-source map	step-to-gold-source map	progress-to-next-source map
$B_{\ell h}$, contrast	$T_{\ell h}$, $D_{\ell h}$	$U_{\ell h}$, stop score
Z_D /answer intervention	identity/coverage intervention	next/repeat/stop intervention
For each model and role, show a small number of real locked-test heatmaps beside raw ablation damage, single-donor rescue, and matched controls.		

Figure 3: Planned component figure. Heatmaps describe candidate access patterns; held-out head-output interventions determine function. Qwen and Gemma head indices need not match.

5.3 From heatmap to functional component

Every role follows one certification ladder:

$$\text{discover} \longrightarrow \text{freeze} \longrightarrow \text{held-out selectivity} \longrightarrow \text{local ablation} \longrightarrow \text{clean patch rescue.} \quad (24)$$

For an endpoint m , normalized recovery

$$\text{Rec} = \frac{m_{\text{patch}} - m_{\text{corrupt}}}{m_{\text{clean}} - m_{\text{corrupt}}}$$

is reported only when the clean-corrupt denominator passes a validation-frozen gate; all raw margins are always shown.

The intervention must first move the role’s proximal semantic endpoint. Broad candidates must affect terminal total quality. Targeted candidates must first affect gold/emitted source identity or unique coverage. Progress-transition candidates must first affect the next source, repetition, or stopping. Pattern patches test localization; source-value and head-output patches test information transport; residual patches localize no more finely than the block. MLP output is tested separately after a block-level state effect is established.

TODO. H1—frozen role screen. On discovery families, compute $B_{\ell h}$, $C_{\ell h}^B$, $T_{\ell h}$, $D_{\ell h}$, and $U_{\ell h}$ for every layer/head in both models using step-to-variant-to-family aggregation. Freeze no more than five candidates per role/model and all thresholds on validation families. Use a family-wise max-statistic screen. On locked families, report raw span mass, density and uniform-baseline sensitivity, gold-target versus emitted-target alignment, score uncertainty, and representative heatmaps without post-test reselection.

TODO. H2—functional component tests. For every frozen candidate, run query-local head-output ablation, clean-to-corrupt single-donor patching, and path patching into the registered state. Broad candidates must first affect Z_D or terminal total quality; targeted candidates must first affect source identity/coverage; transition candidates must first affect next source, repetition, or stop. Then measure complete continuation likelihood and free-running exact count/MAE. Report necessity, partial sufficiency, rescue, side effects, and every matched random, same-layer, attention-mass, wrong-row, wrong-source, and sham control on the same complete sequences.

5.4 Cross-model functional replication

Qwen3-8B and Gemma4-E4B share registered variables and causal gates, not head numbers. The full Qwen analysis freezes the workflow; Gemma candidates are then discovered on Gemma discovery families and evaluated on disjoint locked families under the same role definitions. Replication means that the same semantic intervention changes the same proximal and downstream variables even if layers, head indices, or attention pictures differ. If only Qwen supports a complete role, the claim is restricted and Gemma is reported as a boundary case.

6 Why CoT Helps: A Time-Indexed Causal Chain

6.1 The proposed explanatory chain

The central result cannot be “an ablation lowers accuracy.” The proposed mechanism is a sequence of registered mediators:

$$I_{\text{route},k} \longrightarrow \pi_k \longrightarrow \Gamma_k \longrightarrow (\pi_{k+1}, \text{STOP}) \longrightarrow Z \longrightarrow Y_{\text{full}}, \quad (25)$$

with side interventions

$$I_{\text{state},k} \longrightarrow \Gamma_k, \quad I_{\text{terminal}} \longrightarrow Z.$$

π_k is the aligned source occurrence actually retrieved; Γ_k contains semantic progress and any visible/KV ledger; Z is the registered terminal state; and Y_{full} is the complete registered continuation, parsed count, exactness, and absolute error.

Each intervention has a proximal claim gate. Corrupting a targeted-retrieval component at Q_k must first change the probability or identity of the gold next source. Patching P_k must first change the next source, repetition, or stopping. Patching Z may change the full answer without changing coverage and therefore isolates terminal readout. Clean-output rescue is evaluated beside corruption. Associations among attention, state quality, and accuracy are descriptive; causal language is reserved for paired activation interventions and their randomized controls.

The contrast with Direct is computational. Direct may use the shared predicate-conditioned record states through H_{run} , or it may make many record contributions influence a late state through H_{agg} . CoT exposes a recurrent schedule in which each source decision has a local, testable endpoint and its result persists in the generated prefix. The claim is an algorithmic path change only if the corresponding route, progress, and terminal links in [equation \(25\)](#) are all supported.

6.2 Noise has a measurable referent

We do not use “noise” as a synonym for diffuse attention or an unattractive PCA plot. In the main text it refers to one of two quantities:

1. functional retrieval/update error—missed valid occurrences, false positives, repeated retrieval of one occurrence, wrong next source, or premature/late stopping; or
2. nuisance-conditioned state dispersion ν_{state} in [equation \(17\)](#).

Attention entropy, target mass, and selectivity are routing descriptors. Parse, format, and truncation are behavioral failure categories. They are not folded into one latent “noise” variable.

CoT can improve counting by reducing functional retrieval error: targeted queries increase the margin for one unvisited source, while the trace makes visited identities available for duplicate avoidance and stopping. This does not predict immunity to scale. Larger N creates more opportunities for misses, repeats, and stop errors, and longer L may reduce source margins. The shape of this accumulation is estimated from the error decomposition in [equation \(9\)](#); it is not imposed as an independent-step product law.

6.3 Factorial intervention on route, state, and terminal readout

On locked Qwen families, we cross clean/corrupt/rescued levels for the route, progress carrier, and terminal state:

- **Route intervention.** Ablate or wrong-source patch the frozen targeted component at Q_k . First measure gold/emitted identity and unique coverage.
- **State intervention.** Patch a single clean P_k donor into a matched corrupted prefix, or separately transport trace/KV state. First measure next source, repeat, continue, and stop.
- **Terminal intervention.** Patch a single clean Z donor at the same **Total**: landmark. First measure complete numeric-string likelihood and then free-running exact count/MAE.

The decisive pattern is ordered degradation and rescue: route corruption changes source identity/coverage; state corruption changes future retrieval or stopping; the resulting terminal state and full answer change; restoring a downstream clean mediator rescues only its descendants. A terminal patch that restores the answer without restoring coverage is expected and identifies a readout effect rather than a retrieval repair.

For an endpoint m , we report paired raw differences and compute

$$\text{Rec}(m) = \frac{m_{\text{rescue}} - m_{\text{corrupt}}}{m_{\text{clean}} - m_{\text{corrupt}}}$$

only above a validation-frozen denominator gate. Intervals are clustered by stimulus family and paired randomization tests respect the sibling structure. No single normalized recovery ratio replaces the raw semantic endpoints.

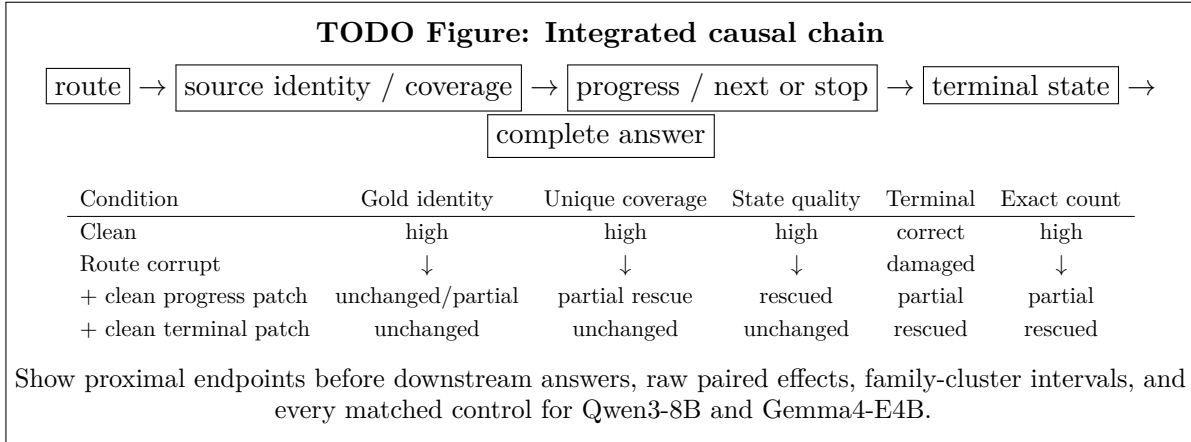


Figure 4: Planned decisive result figure. A causal explanation requires intervention effects to propagate through the registered semantic chain, not merely an accuracy drop.

TODO. M1—integrated Qwen causal chain. On locked Qwen3-8B families, run the route-by-state-by-terminal factorial with clean, corrupt, rescue, and sham levels. For every intervention, report the proximal endpoint first: gold/emitted source identity for route; next source/repeat/stop for progress; complete answer for terminal. Then report unique coverage, false positives, duplicates, sequence likelihood, terminal state quality, exactness, and MAE with family-cluster intervals and paired randomization tests. Include single-donor natural patches, pattern/value/head and residual path decomposition, random/same-layer/wrong-row/wrong-source controls, and free-running validation before and after the first trace error. Do not replace the chain by one accuracy drop or a cross-sectional mediation regression.

6.4 Common-evidence readout and trace-content controls

Retrieval and readout must be separated without pretending that a teacher-forced list is a natural Direct computation. We give both reasoning-policy conditions the same correct, passage-ordered evidence ledger and compare the state at the identical **Total:** token. This *common-evidence readout* intervention fixes unique coverage and is an auxiliary control: if the interface gap disappears, the natural advantage is consistent with retrieval mediation; if it remains, state construction or terminal readout contributes after evidence is fixed.

An equal-length null scratchpad controls for extra tokens and recurrent computation. Counterfactual traces preserve token length and grammar while inserting a matched non-target, a repeated occurrence, or a false stop. A trace-length-matched correct prefix controls for merely having more KV positions. We also compare the same faithful ledger in a user prefix versus an assistant prefix, shuffled faithful evidence, and Direct with a prefilled ledger. These interventions test whether semantic trace content, rather than verbosity, label tokens, or additional compute, drives the benefit.

TODO. M2—controls and Gemma functional replication. Repeat every decisive link in Gemma4-E4B using independently discovered but identically defined functional components. Run the common-evidence readout, equal-length null scratchpad, matched non-target, repeated-source, false-stop, shuffled-ledger, and user-versus-assistant-prefix controls. Claim functional replication if the same semantic links hold even when layers and head identities differ. If only Qwen supports the complete chain, state the model-specific boundary. Claim an improvement beyond coverage only if a fixed-source-set intervention still changes the registered terminal state and the complete answer.

6.5 Claim ladder and falsifiers

Table 6: Evidence required for each level of the paper’s claim. Passing an earlier level does not license a later conclusion.

Claim level	Required evidence	Principal falsifier or weaker conclusion
Behavior	trace-generating interfaces outperform Direct under matched stimuli	effect disappears under provenance, policy, or budget control
Process	source retrieval, duplicates, stopping, and Direct total formation are separately measured	only the final number is available
Representation	grouped held-out decoding and the registered dynamic profile	only token, index, position, or density decodes
Component causality	local semantic damage and clean single-donor rescue on frozen candidates	heatmap/ablation changes without the proximal endpoint
Integrated causality	effects propagate through coverage, progress, terminal state, and answer	null compute matches the gain or downstream restoration fails
Scope	functional chain replicates in both mechanism models and a compatible synthetic setting	implementation or result is checkpoint/template specific

7 Supplementary Synthetic Bridge and Training Dynamics

Synthetic models permit exact source alignment, dense checkpoints, and interventions that are expensive in LLMs. Their role is to develop measurements, expose confounds, and generate predictions. They do not establish that pretrained Qwen or Gemma models use the same circuit.

7.1 What synthetic v10 contributes

Synthetic v10 trains separate 4-layer, 4-head, $d_{\text{model}} = 256$ causal transformers on counts 1–30 with learned absolute positions. Within that setting, it gives five concrete design lessons:

1. final count is linearly readable in both modes, but the centroid path bends with depth, so local or piecewise steering is preferable to one global +1 direction;
2. Direct answer queries contain broad-aggregation candidates, while trace queries contain targeted-retrieval candidates;
3. progress/termination states and final-total states are causally dissociable;
4. natural final-state transplants can transport donor count, whereas some trace-marker states control early close without transporting the final scalar;
5. high trace-readout attention mass does not guarantee that the selected head output carries a total.

These findings motivate the counter-curve, head-role, and mediation experiments above. They also justify the strict separation between representation, routing, and causal transport.

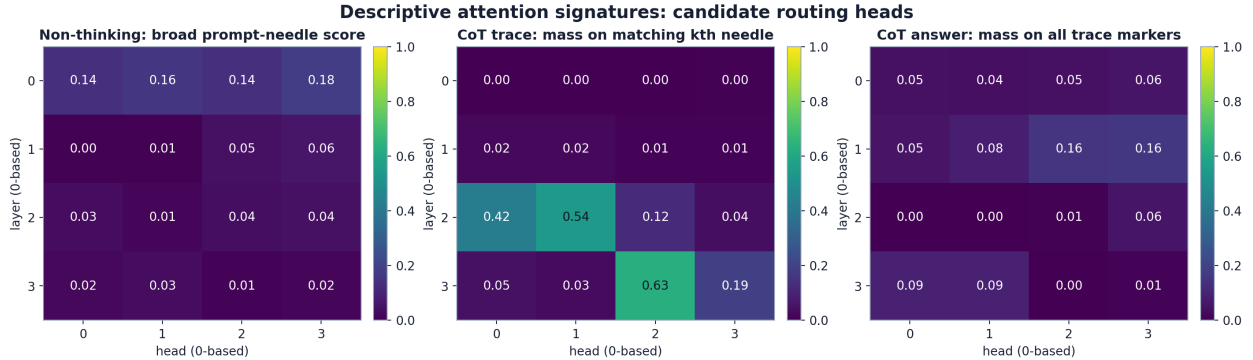


Figure 5: Preliminary synthetic v10 signatures used to design the LLM study. Direct answer queries show broad source access; trace queries show targeted retrieval; trace-to-answer readout is separate. The panels are descriptive until accompanied by query-local ablation and clean patch recovery.

7.2 Why v15 is not the final mechanism setting

Version 15 uses Tiny Shakespeare backgrounds, $d_{\text{model}} = 256$, RoPE or learned relative bias, all-sequence loss, and inserted marker targets. It changes several factors at once and is therefore a design stress test rather than a clean ablation. Final counts can approach ceiling even while whole-trace fidelity falls sharply, and position-only controls at trace/final sites can be highly predictive. The lesson is negative but important: correct final answers do not prove faithful iterative aggregation, and a strong progress probe may decode position or visible step.

We therefore retain v15 only for control development: index permutation, position matching, trace-fidelity decomposition, and separation of progress from final total. It is not used as the primary synthetic claim.

7.3 v20 as the candidate training-dynamics setting

Version 20 counts occurrences of a queried set of three natural characters in a 256-character Tiny Shakespeare window, restricted to totals 1–30. It uses query-first input, RoPE, atomic count tokens, paired data/order for separately trained Direct and trace models, and a two-stage objective: all-sequence loss through step 1500, then task-output loss through step 10,000. Both models have four layers, four heads, and $d_{\text{model}} = 256$.

At the final one-seed checkpoint, the trace model reaches .912 exact count on held-out free-running examples, while the Direct model reaches .335. Whole-trace exactness and ordered-marker accuracy are also high in the trace model. The Direct output error does not exhibit a simple monotone increase with count in that sample, so the proposed state-dispersion analysis remains necessary. Local repair artifacts suggest progress-transition and targeted-retrieval patch effects, including value-dominant transport, but the original run manifest marks part of the causal stage as failed. These results are hypotheses until the stage is rerun from a recorded revision with multiple seeds.

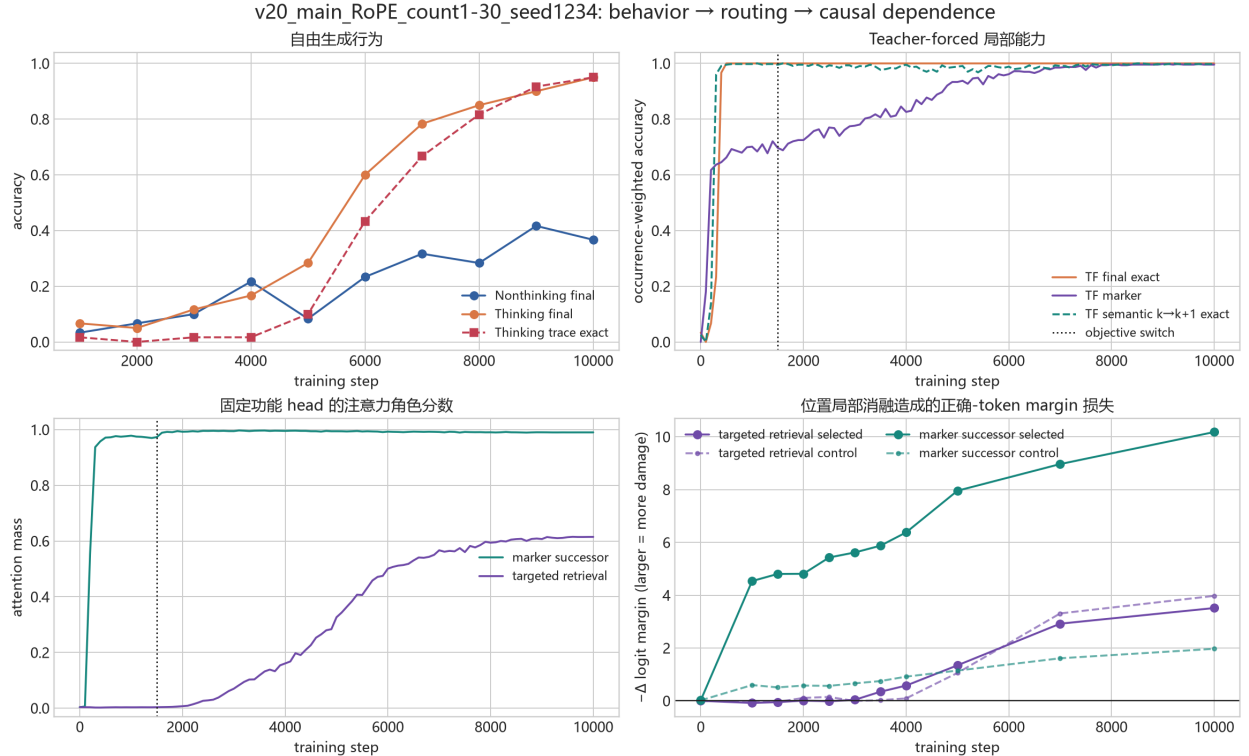


Figure 6: Current one-seed v20 training timeline. Teacher-forced local abilities appear before free-running trace accuracy; progress-transition behavior appears before targeted-retrieval specialization. Because the objective changes at step 1500, this ordering is descriptive until replicated under no-switch and multi-seed controls. Labels are inherited from the internal report.

7.4 Training-dynamics estimands

At saved checkpoints we track free-running exact accuracy, teacher-forced local accuracy, $B_{\ell h}$, $T_{\ell h}$, $D_{\ell h}$, and $U_{\ell h}$ from [section 5](#), held-out count/progress decoding, valid update consistency, state-patch restoration, query-local component ablation, and clean patch recovery. Optimizer step is supplemented by semantic exposure: the cumulative number of supervised occurrences of each source identity, retrieval step, and answer count.

Three outcomes are distinguished:

1. **smooth formation**: continuous mechanism metrics predict behavior without a discrete change point;
2. **replicated transition**: a preregistered change-point model improves held-out likelihood and the ordering recurs across seeds and related causal metrics;
3. **objective/exposure artifact**: the apparent transition tracks the loss switch or event frequency rather than a mechanism-specific phase.

This follows work that uses mechanistic progress measures to explain apparently sudden learning [6, 7].

TODO. S1—controlled multi-seed mechanism dynamics. Freeze one v20-based generator, architecture, tokenizer, checkpoint schedule, and loss mask. Train at least three paired initialization seeds, plus (i) a v15-style all-sequence-throughout control matched in exposure and (ii) an explicit objective-switch control. Save dense checkpoints before and after behavioral onset and every registered objective change. Select each final-checkpoint functional bundle on disjoint discovery families, freeze it, and track the same bundle backward; do not reselect the highest-scoring head at every checkpoint. At each checkpoint compute the same decodable–dynamic–causal ladder and B, T, D, U scores used in the LLM study. Use $N \leq 10$ for direct comparability with the controlled LLM corpus and $N = 11:30$ only as extrapolation. Report seed-level uncertainty and formation order. Keep the dynamics in the main paper only if ordering and causal effects are stable across seeds; otherwise report them as supporting appendix evidence. Any change near the objective switch is an objective intervention, not a spontaneous phase transition.

8 Threats to Causal Identification

Retrieval versus aggregation failure. High list recall with low Direct count accuracy is compatible with aggregation or readout failure, but recall must be measured under the same prompt and decoding. Attention mass can dilute with length, and a multi-hop summary route can lack direct answer-to-source attention.

Visible-index and position leakage. Progress may be decoded from an item number, trace position, or answer distance. Primary anchors occur before source-bearing content or at item end, and analyses cross index permutation, index-free traces, fixed answer position, and variable trace extent.

Capability mismatch and floor effects. A clean CoT manifold and noisy Direct manifold may simply compare success with failure. We use all paired prompts, preregistered overlapping-difficulty

cells, and four correctness strata. Common-correct analyses are explicitly selection-conditioned. If Gemma4-E4B Direct remains at floor, its role is failure-mechanism replication rather than unbiased mediation.

Different compute and decoding. Native CoT uses more generated tokens and may use a different sampling policy. Equal-length null scratchpads, fixed token budgets, deterministic matched decoding, and native-policy reruns separate semantic trace effects from extra compute and policy.

Post-treatment traces. Generated trace style is selected by the model. Canonical teacher forcing establishes local competence; free-running prefixes before the first error establish execution. Neither alone identifies a natural mediation effect.

Attention redundancy and superposition. One role may be distributed across heads, and one head may serve multiple roles. Bundle dose responses, union source coverage, query-local interventions, and clean recovery are preferred to naming one maximally scoring head.

Off-manifold interventions. Large centroid replacements can create unnatural states. Residual-preserving local paths and natural donor transplants are primary; activation norm, nearest-natural-state distance, and downstream KL are reported.

Cross-model non-homology. Qwen and Gemma may implement the same function with different depths and distributed components. Replication is defined by semantic endpoint and causal effect, not matching head indices.

9 Related Work

CoT prompting improves multi-step performance [12], but generated explanations can be unfaithful [9]. Recent work studies information flow and local causal structure in reasoning traces [11, 14]; counting adds exact source alignments and a known algorithmic target. Strong hidden-state probes can be diagnostic without being causal [16].

RULER extends single-needle retrieval to multi-needle and aggregation tasks [4]. Transformer counting can be limited by representation interference, vocabulary, and context length [15]. Our contribution is to compare the routing and state-update mechanisms of direct and trace-based counting in the same task.

The causal tools follow activation patching and causal tracing [5], circuit discovery [1], query-specific circuit analysis [10], and activation steering [8]. We distinguish local ordered-token successor primitives [3] from the semantic progress-transition and continue/stop control required to cover a long-context source set.

10 Limitations

The causal LLM experiments in this draft are registered rather than complete. The existing behavioral audit mixes some checkpoint pairs, decoding policies, and protocol failure modes. Open-

weight Qwen and Gemma results may not generalize to proprietary or non-autoregressive systems. Linear probes can miss nonlinear codes; flexible probes can memorize nuisances. Patching can be off-manifold, and ablation can trigger compensation. Native source alignment is imperfect after the first trace error. Synthetic models differ from LLMs in scale, pretraining, tokenization, and task familiarity. Finally, the response-surface forms were explored on existing data and must earn the name “law” by predicting the untouched Qwen3-14B cells.

11 Reproducibility and Artifact Plan

The final artifact will include request-level data or hashes, exact prompts, checkpoint/tokenizer revisions, chat templates, generation parameters, semantic-anchor parsers, intervention code, discovery/reporting splits, and seed-level tables. Each figure will be generated from a frozen table with a manifest and checksum. Numerical exactness, parse, format, and truncation remain separate. Synthetic checkpoints and optimizer states will be retained at the declared cadence.

TODO. R0—frozen analysis registry. Create one versioned YAML file containing: task unit; model/mode pairs; checkpoint hashes; core and mechanism cells; semantic sites; trace alignment rules; discovery/reporting groups; head scores and bundle selection; count-state and noise estimands; intervention dose grids; endpoint definitions; denominator gates; negative controls; multiplicity correction; minimum examples/seeds; exclusion codes; and the claim ladder in table 6. Generate every table caption’s sample size and provenance tag from this registry.

12 Conclusion

This paper is designed to answer “How does CoT help language models count?” with a causal comparison of computations rather than a catalog of probes and attention maps. Direct answering is tested for a record-wise running update and for terminal broad aggregation. CoT is tested for targeted retrieval, trace writing, semantic progress transition, next-source or stop control, and terminal readout. The strongest conclusion requires interventions to propagate through source identity, unique coverage, progress and stopping, terminal state, and the complete count.

The claim will be narrowed to the links the data actually support. A decodable Direct record state without the sibling-controlled +1/0 profile and propagated patches is ordinal information, not a running counter. A CoT state that fails label controls is numeric scaffolding, not semantic progress. An attention component without a proximal source or stopping effect is a signature, not a circuit. If equal-length null scratchpads reproduce the benefit, prefer an extra-compute account; if common evidence removes the mode gap, prefer retrieval mediation; if a residual gap remains, test state construction and terminal readout. Qwen3-8B and Gemma4-E4B provide functional replication only when the same semantic links hold, even if their layers and head identities differ.

References

- [1] Arthur Conmy, Augustine N. Mavor-Parker, Aengus Lynch, Stefan Heimersheim, and Adrià Garriga-Alonso. Towards automated circuit discovery for mechanistic interpretability. *Advances in Neural Information Processing Systems*, 2023.
- [2] Gemma Team, Sherif El Abd, Vaibhav Aggarwal, Robin Algayres, Alek Andreev, Olivier Bachem, et al. Gemma 4 technical report. *arXiv preprint arXiv:2607.02770*, 2026.
- [3] Rhys Gould, Euan Ong, George Ogden, and Arthur Conmy. Successor heads: Recurring, interpretable attention heads in the wild. *arXiv preprint arXiv:2312.09230*, 2023.
- [4] Cheng-Ping Hsieh, Simeng Sun, Samuel Kriman, Shantanu Acharya, Dima Rekesh, Fei Jia, Yang Zhang, and Boris Ginsburg. RULER: What’s the real context size of your long-context language models? *Conference on Language Modeling*, 2024.
- [5] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. *Advances in Neural Information Processing Systems*, 2022.
- [6] Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. *International Conference on Learning Representations*, 2023.
- [7] Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas Joseph, Nova DasSarma, Tom Henighan, et al. In-context learning and induction heads. *Transformer Circuits Thread*, 2022.
- [8] Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Steering language models with activation engineering. *arXiv preprint arXiv:2308.10248*, 2023.
- [9] Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. Language models don’t always say what they think: Unfaithful explanations in chain-of-thought prompting. *Advances in Neural Information Processing Systems*, 2023.
- [10] Kevin Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt. Interpretability in the wild: A circuit for indirect object identification in GPT-2 Small. *International Conference on Learning Representations*, 2023.
- [11] Dennis Wei, Yannis Belkhir, Erik Miehling, and Radu Marinescu. Local causal attribution of chain-of-thought reasoning. *Mechanistic Interpretability Workshop at ICML*, 2026.
- [12] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc V. Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems*, 2022.
- [13] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.
- [14] Hao Yang, Qinghua Zhao, Lei Li, Lingyi Meng, and Mengda Yu. How chain-of-thought works? tracing information flow from decoding, projection, and activation. *Annual Meeting of the Association for Computational Linguistics*, 2026.

- [15] Gilad Yehudai, Haim Kaplan, Guy Dar, Royi Rassin, Asma Ghandeharioun, Mor Geva, and Amir Globerson. When can transformers count to n ? *arXiv preprint arXiv:2407.15160*, 2024. Version 3, February 2026.
- [16] Aojie Yuan, Zhiyuan Julian Su, Haiyue Zhang, Yi Nian, and Yue Zhao. Hidden error awareness in chain-of-thought reasoning: The signal is diagnostic, not causal. *Mechanistic Interpretability Workshop at ICML*, 2026.

A Complete Behavioral Tables and Model Exclusions

A.1 Requested response-surface panel on the full grid

Table 7: Preliminary numerical exact accuracy over the full $10 \times 5 \times 10$ grid (500 requests per available model-mode cell). These descriptive values precede the matched-policy rerun.

Model / logical slot	Direct	Index	Bullet	CoT
Qwen3-4B	.432	.888	.806	.928
Qwen3-8B	.570	.962	.894	.956
Qwen3-14B	<i>not yet evaluated</i>			
Qwen3-32B	.600	.974	.858	.976
Gemma4-E4B	.248	.882	.796	.832
GLM-4 / GLM-Z1 [†]	.302	.710	.636	.684
Cogito-v1-Preview-8B	.376	.516	.776	.720
Nemotron-Nano-v2-9B	.236	.666	.580	.636
Nemotron-3-Nano-4B	.366	.652	.546	.654

[†]The GLM row stitches three modes from GLM-4-9B and CoT from GLM-Z1-9B; it is excluded from same-checkpoint mode effects. Gemma4-E4B Direct numerical exactness (.248) differs from registered success (.240); native CoT numerical exactness (.832) differs from registered success (.792).

A.2 Models excluded from the main panel

Models are not removed because they disagree with the hypothesis. They are excluded from the main response-surface or mechanism panel when a frozen evaluation gate fails: incomplete mode coverage, a different checkpoint for the reasoning condition, severe parse/format/truncation confounding, or accuracy too close to floor for the planned causal contrast. Their numerical accuracy remains visible.

Table 8: Other audited models and why they are assigned to the appendix. Entries are full-grid numerical exact accuracy; a dash indicates that the mode was not run on that checkpoint.

Model / checkpoint setup	Direct	Index	Bullet	CoT	Main-panel exclusion reason
Qwen3-1.7B	.196	.288	.396	.602	Low-capability floor; enumeration also has substantial parse/truncation failures. Retained as a phase-boundary sensitivity.
Gemma4-12B	.514	.950	.746	.908	Native numerical accuracy is high but registered success is .314 because of final-channel format behavior. Not selected for the co-primary 8B-scale mechanism comparison.
DeepSeek-R1-0528-Qwen3-8B	–	–	–	.672	Native-only cell; no within-checkpoint mode quartet.
Granite-3.3-Instruct-8B	.394	.440	.374	.440	Low and weakly separated modes plus format/protocol failures; unsuitable for non-floor mechanism mediation.
Minstral Instruct / Reasoning	.658	.892	.756	.770	CoT uses a separate reasoning checkpoint; native registered success is .020 despite .770 numerical exactness, creating a severe format confound.
OLMo3 Instruct / Think	.234	.238	.302	.508	CoT uses a separate checkpoint; Native parse is .746 and truncation .254, while Direct is near floor.

The two Nemotron models remain in the requested cross-family panel, but their full per-model Index/Bullet surfaces are appendix material because enumeration has nontrivial parse/truncation failures. Nemotron-3 Native is comparatively clean. Cogito Index likewise requires parser/protocol repair before conditional signed-bias fits can be interpreted.

B Registered Experiment Ledger

Table 9: Unified TODO ledger. The required output is part of the preregistration, not a suggested visualization.

ID	Experiment	Required output and controls	Claim enabled
B0	Configuration and failure audit	Frozen revisions/templates/budgets/parsers; request-level provenance, parse, truncation, protocol, and leakage flags	Comparable behavioral panel
B1	Paired behavior grid	Four modes on 500 frozen stimuli; full requested model panel; exact count, registered success, error budget, and family-cluster intervals	Robust behavioral phenomenon
B2	Held-out response surfaces	Qwen within-family development; untouched 14B prediction; grouped held-out scoring and coefficient stability	Optional empirical law
D1	Controlled mechanism corpus	Tokenizer-specific equal-length siblings; stratified slots; family-disjoint splits; audited spans and invariant delimiters	Identifiable representation and patching tests
D2	Prompt-policy controls	2×2 instruction by reasoning switch; predicate/policy placement; prefix hashes; paired decoding seeds; null scratchpad	Controlled algorithmic contrast
R1	Registered-state decoding	Family-weighted held-out probes; nuisance controls; individual-state PCA; conditional dispersion and error strata	Count/progress representation
R2	Direct dynamics and patches	Prefix-shift 0/1/0; total shift; same-position unchanged donors; predicate-after negative control; terminal tests	H_{run} versus H_{agg}
R3	CoT anti-leakage and state control	No-index/random/permuted labels; new/non-target/repeated prefixes; residual versus trace/KV transport; steering and natural patches	Semantic progress carrier
H1	Frozen role screen	B, C^B, T, D, U ; discovery/validation/test split; family-wise max-statistic; raw and conditional mass	Candidate functional components
H2	Functional component tests	Query-local ablation; single-donor clean patch; path patch; semantic proximal endpoints and matched controls	Necessity and partial sufficiency
M1	Integrated Qwen causal chain	Route/state/terminal factorial; coverage, progress, terminal, and complete answer endpoints	Mechanism-to-behavior link
M2	Controls and Gemma replication	Common evidence; null scratchpad; semantic corruptions; independently discovered co-primary replication	CoT explanation and scope
S1	Synthetic emergence	Frozen v20 basis; at least three seeds; all-sequence and objective-switch controls; fixed bundle tracked across checkpoints	Developmental support
R0	Reproducible analysis registry	Versioned YAML plus manifests/checksums	All figures and tables reconstruct from frozen inputs and declared exclusions

For the Qwen mechanism claim, B0–B1, D1–D2, R1–R3, H1–H2, and M1 are required. M2 is additionally required for the stated Qwen–Gemma functional replication claim. B2 and S1 determine optional law and developmental scope; R0 is required for every released result.

C Prompt Serialization and Activation Manifest

The frozen V2 user message is serialized in this order:

You will need to count all city-score audit records
in the passage below.
A city-score audit record names one city and gives
that city’s numeric score.
<passage>
[byte-identical frozen passage]
</passage>
How many city-score audit records are in the passage?
[mode-specific response-policy block]

The response-policy block is Direct, Bullet, Index, or Native CoT as specified in [table 1](#). Mechanism contrasts store both raw message content and the output of the exact model-specific chat template; rendered text and token IDs, rather than the unrendered message alone, determine prefix identity. Each activation row stores `family_id`, model and tokenizer revisions, interface, predicate/policy placement, raw messages, rendered-prefix hash, complete input and continuation token IDs, source character spans, model-specific source token spans, registered landmark indices, data seed, decoding seed, `source_version`, hook name, and capture dtype. Assertions verify byte-identical passages across interfaces, identical rendered token prefixes whenever a shared-prefix claim is made, equal sequence length for each clean/corrupt pair, and no truncation before a registered endpoint.

D Predicate and Policy Placement Matrix

Table 10: The causal mask and prompt serialization determine which state comparisons are identifiable. Only the first row belongs to the frozen behavioral grid; the remaining rows are mechanism controls.

Condition	Passage-time information	Valid use
Predicate-before, suffix-after	Predicate known; response suffix unseen; reasoning policy also unseen only when the rendered-prefix audit passes	Shared H_i candidate when prefix identity passes; otherwise analyze within interface; mode-specific Z_D, P_k remain valid
Predicate-after	Task predicate unseen	Content/position control at H_i ; terminal tests remain valid
Predicate-repeat	Predicate known and refreshed after passage	Separates passage-time availability from instruction recency
Policy-before	Predicate and response policy known	Tests interface-specific passage encoding; mechanism only

E Planned Result Figures and Reproducibility Checklist

The planned empirical figures are:

1. behavior surfaces and the complete failure budget (TODO B1–B2);
2. registered-state geometry, dynamic updates, and natural-state patches (TODO D1–D2, R1–R3);
3. real attention maps beside held-out ablation and patch effects for the three component roles (TODO H1–H2);

4. intervention-to-coverage-to-progress-to-answer effects, including the common-evidence control (TODO M1–M2); and
5. synthetic emergence order with seed uncertainty (TODO S1).

The final artifact freezes model and tokenizer files, chat templates, generation parameters, separate data and decoding seeds, stimulus and rendered-prefix hashes, parser and alignment code, character-to-token maps, landmark definitions, hook locations, intervention code, checkpoint IDs, and all discovery/validation/test splits. Request-level outputs are retained rather than only aggregates. Hardware, precision, batching, library versions, and synthetic generator provenance are reported.

F Prompt, Trace, and Intervention Controls

- Direct total-only; numbered list; bullets; index-free item tags; native CoT; equal-length semantically null scratchpad.
- Same faithful list in user versus assistant prefix; model-generated list replayed under teacher forcing; Direct with a prefilled ledger.
- Correct identity with wrong attribute; wrong identity with correct attribute; missing, invalid, and same-template distractor records. A second independent valid input occurrence still contributes +1; only repeated retrieval of the same occurrence ID is a CoT progress no-op.
- Candidate ordinal fixed while validity flips; number of seen candidates fixed while valid-prefix count changes.
- Source order permutation, visible-index permutation, symbolic indices, and item anchors before versus after visible numerals.
- Fixed trace extent with different unique coverage; fixed coverage with different duplicate counts; fixed answer position with variable trace extent.
- Fixed (N, L) with dispersed versus clustered targets and early/middle/late source-depth strata.
- Same final total with different source sets and traces; same trace length with different totals.
- Query-local mean/resample ablation and zero-ablation sensitivity; primary rescue with one matched natural donor; score-matched, random, and wrong-layer bundles; wrong query rows and source donors; norm-matched subspaces.
- Complete parsed numeric-string probability and free-running exact count, never only a first-digit logit.

G Additional Statistical Specification for State Dispersion

For interface m , registered site s , layer ℓ , and semantic label c , fit a family-weighted decoder on discovery families only. On locked families define decoder bias, variance, and mean squared error:

$$b_{ms\ell}(c) = \mathbb{E}[\hat{c} - c \mid c], \quad v_{ms\ell}(c) = \text{Var}(\hat{c} - c \mid c), \quad q_{ms\ell}(c) = v_{ms\ell}(c) + b_{ms\ell}(c)^2. \quad (26)$$

Decoder error and ν_{state} in [equation \(17\)](#) answer different questions and are reported separately. The former measures accessibility to a chosen decoder; the latter measures within-label dispersion relative to adjacent separation after nuisance residualization.

On the controlled $N \in \{0, \dots, 6, 8, 10\}$ corpus, estimate prespecified slopes of $\log(\nu_{\text{state}} + \epsilon)$, $\log(v + \epsilon)$, and $\log(q + \epsilon)$ against count or semantic progress while controlling for context length, depth, terminal position, trace extent, template, and correctness stratum. Nonadjacent sparse values use their observed Δc and are never treated as missing adjacent counts. Family-cluster intervals resample

families. The fair interface comparison is Direct Z_D versus controlled CoT Z_C^{ctrl} ; H_i versus P_k is cross-role and exploratory. A larger-count-noise claim requires convergent separation, dispersion, decoder-error, and causal-transport evidence rather than one significant slope.

H Architecture-Specific Intervention Notes

Both co-primary models use grouped-query attention, so an intervention must specify whether it acts on a query head, a shared key/value head, or the concatenated pre-output-projection head slice. For each layer, the artifact records tensor shape, rotary position convention, attention backend, and hook location. The causal decomposition is:

attention pattern $\rightarrow$ source-weighted value $\rightarrow$ query-head output before o_{proj}
 $\rightarrow$ attention block output $\rightarrow$ post-layer residual.

Pattern-only recovery supports routing; value/head-output recovery supports information transport; residual-only recovery localizes the effect no more finely than the layer. MLP output is patched separately only after a block-level state effect is established. Cross-mode and cross-model activation patches are exploratory because the residual bases and manifolds need not align.

I Claim-to-Evidence Reporting Rules

1. Every exact number in the main text must trace to a frozen CSV/table and a manifest, not only to HTML prose.
2. PCA and centroids describe geometry. “Counter” additionally requires held-out decodability, the registered +1/0 dynamic test, and downstream transport.
3. “Broad aggregation,” “targeted retrieval,” and “progress transition” remain candidate labels until independent query-local ablation and single-donor clean restoration establish the proximal semantic endpoint.
4. Teacher-forced effects establish local competence. Natural execution requires the same signed effect on free-running prefixes before the first error.
5. Synthetic results establish measurement feasibility, not the mechanism of a pretrained LLM.
6. A response surface becomes a law only after untouched-model prediction. Failed validation remains in the paper as a negative result.